Ben-Gurion University of the Negev
Faculty of Engineering Science
School of Electrical and Computer Engineering
Dept. of Electrical and Computer Engineering

Fourth Year Engineering Project
Progress Report

Project name - Geometric shaping in the FSO turbulent channel

Project number:	p-2026-093
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Supervisors:	Prof Derevyanko Stanislav
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1. Project Objective:

The objective of this project is to maximize the theoretical channel capacity, measured as Average Mutual Information (AMI), of a Free-Space Optical (FSO) communication link operating under weak atmospheric turbulence and Additive White Gaussian Noise (AWGN). This is achieved by applying Geometric Shaping (GS) to an intensity-modulated (IM/DD) M-PAM constellation while strictly enforcing physical system constraints: non-negativity ($x \geq 0$) and average transmit power.

2. Work Completed to Date:

Task/Milestone	Progress Status	Current Outcome
Theoretical formulation and scope definition	Completed	Channel assumptions, constraints, deliverables, risks, and test methodology were formalized in the preliminary report.
Optimization engine	Completed	A Simulated Annealing framework with parallel multi-start execution, and power-constraint enforcement was implemented in MATLAB.
AMI evaluation pipeline	Completed	A Dual-Evaluator Architecture was developed: a fast Gauss-Hermite (GH) quadrature for the SA loop, and an independent high-precision adaptive integral() for validation.
Channel-level verification	Completed	Conditional densities, a-posteriori probabilities, decision boundaries, SER, and AMI consistency tests were executed successfully.
Performance simulations	Completed	Grid sweeps over SNR and turbulence scenarios were performed for M=8 and compared against uniform PAM.

3. Main Achievements and Current Results:

Based on the recent simulation runs and validation tests, the project is advancing successfully and has already met and exceeded its primary milestones. The core achievements are detailed below:

3.1. Accurate FSO Channel Modeling and Validation:

- We successfully implemented the complete log-normal FSO channel model, capturing both the multiplicative turbulence fading and additive white Gaussian noise.
- To overcome numerical instability, we applied a variable change ($t = \ln h$) and bounded integration limits, ensuring highly stable $P(y|x)$ evaluations.
- Validation scripts confirm our model's precision, demonstrating that the A Posteriori Probability (APP) normalization error is negligible (on the order of 10^{-16}).

3.2. Efficient Simulated Annealing and Evaluator Accuracy:

- The Simulated Annealing (SA) optimization engine is fully operational and highly efficient. By implementing a Dual-Evaluator Architecture a fast Gauss-Hermite (GH) estimator for the SA loop and a high-precision adaptive integral() for validation we eliminated severe computational bottlenecks. For context, at M=8 and SNR=20 dB, executing 6 parallel starts with 50,000 iterations per start, the algorithm completely optimizes the constellation for a given turbulence level in roughly 4 minutes on a standard 8-core desktop processor (Intel Core i7-10700, 16GB RAM).

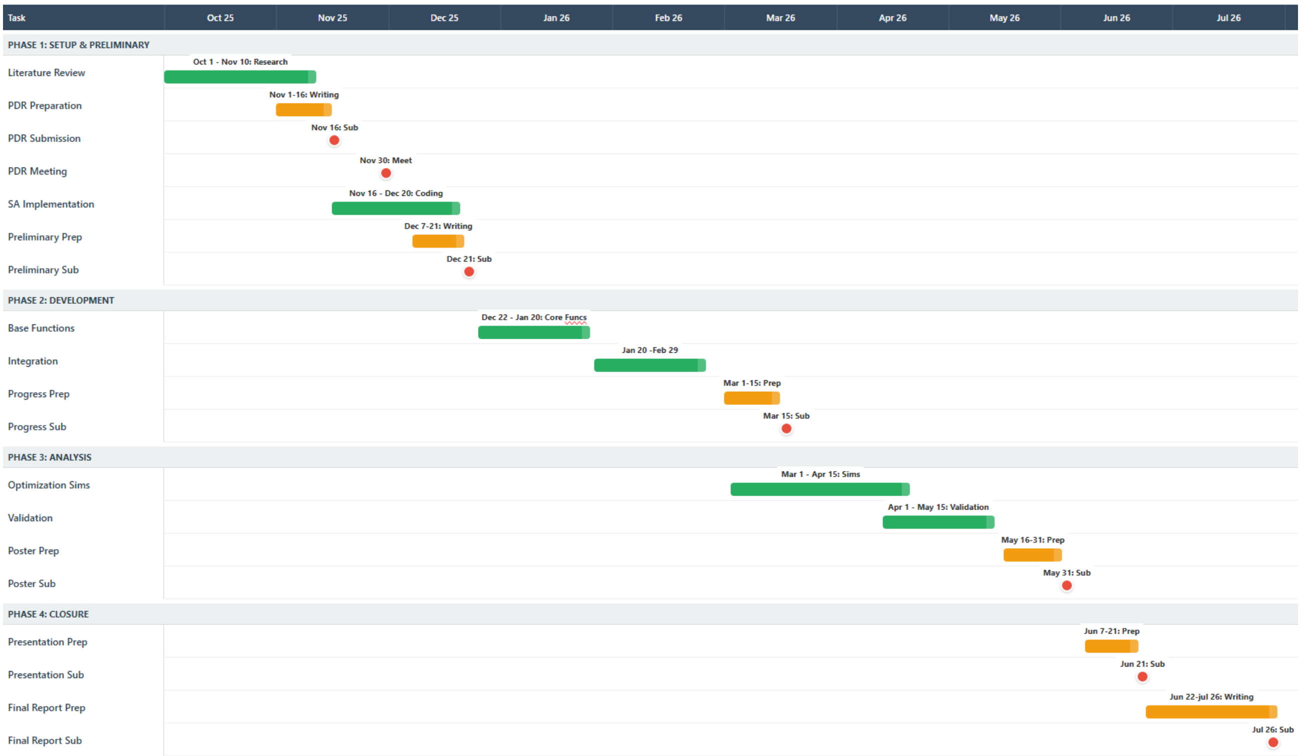
- 3.3. Substantial Shaping Gains Under Turbulence:
  The Geometric Shaping methodology consistently outperforms standard Uniform-PAM across various turbulence levels and SNRs. For an M=8 constellation at an SNR of 20 dB, the validated shaping gains under different Rytov variances σ_R^2 are:

σ_R^2	PAM AMI	OPT AMI	**Gain (bits)**
0.00	2.524	2.602	**+0.078**
0.10	1.315	1.676	**+0.361**
0.20	1.067	1.406	**+0.339**
0.30	0.932	1.280	**+0.348**

- Furthermore, a comprehensive SNR sweep (ranging from 5 dB to 30 dB) confirms that GS constellations consistently outperform uniform PAM across all SNR and turbulence levels, achieving a robust shaping gain of 0.32–0.49 bits/sym. This proves the stability of the GS approach, showing that the optimized constellation geometries effectively mitigate turbulence-induced fading regardless of the baseline AWGN level.
- The constellation comparison plots reveal the physical intuition behind the SA algorithm's success: as turbulence increases, the algorithm intelligently abandons the uniform spacing and clusters the transmission symbols at lower intensity levels to avoid the severe, signal-dependent multiplicative noise found at higher amplitudes.

4. Compliance with Tasks and Milestones:

The current project state is aligned with the planned progress stage. The main software components defined in the preliminary report have already been implemented, integrated, and tested. In particular, the optimization algorithm, channel model, independent verification path, and comparative performance simulations are all available in working form. At this stage, the project has moved from planning and formulation into validated experimentation and quantitative performance analysis.



5. Remaining Work:

Based on our progress and the current results, the remaining project tasks are defined as follows:

- **Higher-Order Constellations:** Extend the SA optimization framework to higher-order M-PAM constellations (e.g., M=16, 32) to analyze how the shaping gain scales with denser symbol packing in the non-convex AMI landscape.
- **Robustness Analysis:** Investigate the robustness of our optimized constellations to channel parameter mismatch, specifically analyzing performance degradation under Rytov variance σ_R^2 estimation errors at the receiver.
- **Peak-Power Constraints:** Incorporate strict peak-intensity limits into the optimization constraints (which is crucial for physical FSO eye-safety regulations) and evaluate shaping geometries under joint average and peak power constraints.
- **Final Deliverables:** Consolidate the database of optimized configurations and complete the writing of the final project report and final presentation.

6. Summary

In summary, the project has successfully achieved its primary technical milestone: the development of a robust, fully validated Geometric Shaping optimization engine for the FSO turbulent channel. By utilizing a dual-evaluator architecture and rigorous statistical validation tests, we eliminated numerical instabilities and verified our channel model's precision.

The current results demonstrate that our optimized GS constellations consistently and significantly outperform standard uniform-PAM across a wide range of SNRs (5–30 dB) and turbulence levels, yielding robust shaping gains of 0.32–0.49 bits/symbol. The project is progressing according to plan, having transitioned from the theoretical and architectural design phases into advanced analysis, producing quantitatively and meaningful research outcomes.

המלצת ציון לדו"ח התקדמות:

מספר הפרויקט: P-2026-093 **שם הפרויקט:** Geometric shaping in the FSO turbulent channel.
שם המנחה פרופ' דרביאנקו סטניסלב.
שמות הסטודנטים: ישי דולב, שלמה עדי.

%	קריטריון	1 - חלש	2 - בינוני	3 - טוב	4 - ט"מ	5 - מצוין
14	הגדרת המטרה - האם מטרת הפרויקט ברורה? המטרה צריכה לכלול גם את התרומה הצפויה מהשלמת הפרויקט.					
14	ארכיטקטורת הפתרון - האם בוצעה סקירה טובה של הנושא? האם הובהר מהי הארכיטקטורה המוצעת לפתרון ומדוע היא מתאימה לפתרון הבעיה? האם הייתה התייחסות לאלטרנטיבות?					
14	הצגת התוצאות הראשוניות - האם הוסברו המשמעויות של התוצאות (החלקיות), או המסקנות הנובעות מהן, האם רק הייתה הצגה עובדתית של תוצאות?					
15	ההצגה עצמה - האם יש מבנה הגיוני להצגה? מבוא, רקע ו-SoTA, מטרות, שיטה, פתרון, תוצאות, מסקנות.					
14	רמת קושי - כיצד הנך מעריך את עבודת הסטודנט על הפרויקט עד כה, ביחס לסטודנט *ממוצע* בקורסים אותם אתה מלמד?					
15	הבנה ועצמאות - האם נראה שהסטודנט מבין את העבודה שעשו, האם המענה לשאלות מספק? האם זו הייתה עבודה עצמאית?					
14	שורה תחתונה - כיצד הנך מעריך את התקדמות הפרויקט ביחס לכלל הפרויקטים שראית?					

הערכת רמת הקושי של הפרויקט: קל מאוד / קל / בינוני / קשה / קשה מאוד

הערות:______________________________________